

Daily Screen Time and Calorie Intake Analysis

Abstract

This study examines the relationship between daily smartphone screen time and total caloric intake using 26 days of self-collected data. Exploratory Data Analysis (EDA) and machine learning models were applied to determine whether increased screen exposure is associated with higher caloric consumption. The results indicate a strong positive correlation ($r = 0.9550$, $p\text{-value} = 3.66 \times 10^{-14}$), suggesting that increased screen time aligns with increased calorie intake. Linear Regression achieved an R^2 score of 0.8655, demonstrating strong predictive capability.

1. Introduction

Smartphone usage has become an integral part of modern life, often occupying several hours of an individual's daily routine. Extended periods of screen exposure may contribute to behavioral consequences, such as distracted eating or reduced physical activity. This project investigates whether daily screen time has a measurable effect on total calorie consumption using statistical analysis and machine learning techniques.

2. Motivation

Digital behaviors—especially prolonged smartphone use—may be associated with unhealthy eating patterns. Individuals may consume additional calories due to snacking while using their phones, reduced physical activity, or irregular meal timings. To validate these assumptions from a data-driven perspective, I recorded my own daily screen time and caloric intake over a 26-day period and analyzed the relationship using Python-based tools.

3. Data Description

The dataset consists of 26 consecutive days of observations. Three variables were recorded:

- Date
– Observation day (YYYY-MM-DD)
- Screen_Time – Daily smartphone use (Minutes)
- Total_Calories – Daily caloric intake (kcal)

4. Methodology

The analysis followed the standard data science pipeline: 1. Data Cleaning (checking missing values, structuring, sorting) 2. Exploratory Data Analysis (descriptive statistics, histograms, scatter plot, correlation analysis) 3. Machine Learning (Linear Regression and Random Forest Regressor using an 80/20 train-test split)

5. Results

5.1 Exploratory Data Analysis (EDA) Key statistical findings: • Number of days: 26 • Mean screen time: 560.58 minutes/day • Mean calorie intake: 3025 kcal/day • Correlation (r): 0.9550 • P-value: 3.66×10^{-14} Interpretation: There is a very strong and statistically significant positive correlation between daily screen time and caloric intake. As screen time increases, caloric intake also tends to rise.

5.2 Machine Learning Performance

Linear Regression: • MSE: 14651 • R^2 Score: 0.8655 Random Forest Regressor: • MSE: 27708 • R^2 Score: 0.7455 Linear Regression performed better due to the strong linear relationship and the limited dataset size.

6. Discussion

The strong correlation between screen time and caloric intake highlights a meaningful relationship. Even without additional variables such as sleep or exercise, screen time alone provides substantial predictive power. These findings suggest that digital habits influence personal health behaviors and should be considered in lifestyle analysis.

7. Limitations

This project has several limitations: • Dataset includes only 26 days • Data from one individual only • Calorie values are manually entered and may contain inaccuracies • Additional lifestyle variables were not included

8. Future Work

Possible improvements include: • Collecting a larger dataset over a longer period • Adding variables such as sleep, workout duration, step count, and macronutrients • Testing models such as Support Vector Regression or time-series forecasting • Developing an interactive dashboard

9. Conclusion

This study concludes that daily smartphone screen time and calorie intake are strongly and significantly correlated. Higher screen time coincides with increased caloric consumption. With more data and additional variables, predictive accuracy could be improved, offering deeper insights into the connection between digital habits and nutritional behavior.

10. References

• Digital Wellbeing (Android) • MyFitnessPal • Scikit-learn documentation • Pandas & NumPy documentation